EE 445/645: Physical Models in Remote Sensing (Spring 2026)

Chapter 02-Part 03: Leaf Optics – Notes (See C2-P3 PPTs)

Prof. Ranga B. Myneni, Boston University

ranga.myneni@gmail.com

§1. Introduction

The interaction of electromagnetic radiation with plant leaves constitutes one of the fundamental problems in vegetation remote sensing. All information about the biochemical composition and physiological state of vegetation that can be extracted from remotely sensed measurements of reflected or transmitted radiation must ultimately be traceable to the optical properties of individual leaves and their aggregates. It is therefore essential to develop a rigorous understanding of how photons interact with leaf tissues before proceeding to the canopy-level radiative transfer problems that are the ultimate concern of remote sensing.

In this chapter we shall examine the optical properties of plant leaves, proceeding from the morphological forms and internal structures that govern light–matter interactions, through the biochemical constituents responsible for spectral absorption, to the mathematical models that describe how leaves scatter radiation. We treat two leaf forms that between them encompass the great majority of terrestrial vegetation: the flat broadleaf, characteristic of herbaceous plants, crops, and broadleaved trees; and the conifer shoot, the functional scattering unit of needleleaf forests. The former presents a relatively simple planar geometry amenable to direct application of the classical Fresnel and Lambert theories; the latter introduces the additional complexity of a three-dimensional body composed of multiple cylindrical needles arrayed around a central axis, requiring the apparatus of recollision probability and ensemble-averaged specular reflection.

The chapter is organised as follows. Sections 2 and 3 describe leaf forms, internal structure, biochemical constituents, and their absorption spectra. Section 4 distinguishes the two classes of light–leaf interaction—surface reflection and interior scattering—that motivate the mathematical framework of the subsequent sections. Section 5 defines the leaf scattering phase function, albedo, and hemispherical reflectance and transmittance. Section 6 decomposes the scattering phase function into diffuse and specular components. Sections 7 and 8 present two models for the diffuse component: the PROSPECT radiative transfer model and the bi-Lambertian approximation. Section 9 develops the Cook–Torrance microfacet model for the specular component and establishes its normalisation. Section 10 extends the theory to conifer shoots, treating both the diffuse (bi-Lambertian) and specular (Cook–Torrance) components within the recollision probability framework. Section 11 compares broadleaf and conifer shoot scattering properties. Section 12 offers concluding remarks.

§2. Leaf forms

Two leaf forms concern us in the radiative transfer theory of vegetation canopies. The first is the flat broadleaf, a thin laminar organ typically 0.1–0.5 mm in thickness, with lateral dimensions of several centimetres. Broadleaves are the predominant foliar type in herbaceous vegetation (grasses, shrubs, and crops) and in broadleaved forests and woodlands. The second is the conifer needle, a narrow elongated organ of roughly cylindrical cross-section, typically 1–3 mm in width and 20–100 mm in length. In needleleaf forests, individual needles are clustered into shoots—the functional scattering units that interact with radiation at the canopy level. Forests may be purely broadleaved, purely needleleaved, or mixed; savannas typically combine both forms.

The distinction between broadleaf and shoot is fundamental because the angular distribution of scattered radiation differs profoundly between the two forms. A flat broadleaf presents a single macroscopic surface normal about which specular reflection is concentrated. A conifer shoot, by contrast, exposes a continuum of needle surface normals arrayed around the twig axis, distributing specular reflection over a far broader angular range. This geometric difference propagates through the entire radiative transfer hierarchy, from the leaf scattering phase function to the bidirectional reflectance of the canopy.

§3. Leaf structure, biochemical content, and absorption spectra

The cross-section of a broadleaf reveals a layered internal architecture that governs the transport of radiation within the leaf (Fig. 1). The outermost layer is the epidermis, a single layer of flattened cells covered by the cuticle—a thin waxy film, typically 0.1–2 μm thick, composed principally of cutin and epicuticular waxes. The cuticle is the surface at which specular reflection occurs, and its roughness determines the angular width of the specular lobe. Below the adaxial (upper) epidermis lies the palisade mesophyll, a densely packed layer of vertically oriented cylindrical cells containing chloroplasts. Beneath this is the spongy mesophyll, characterised by loosely arranged cells separated by abundant intercellular air spaces. These air–cell wall interfaces are the principal sites of internal scattering: the refractive index discontinuity between cell walls ($n \approx 1.4$–1.5) and air ($n = 1.0$) causes partial reflection and refraction at each interface, randomising the direction of photons that have entered the leaf interior. The abaxial (lower) epidermis completes the structure and bears most of the stomatal openings.

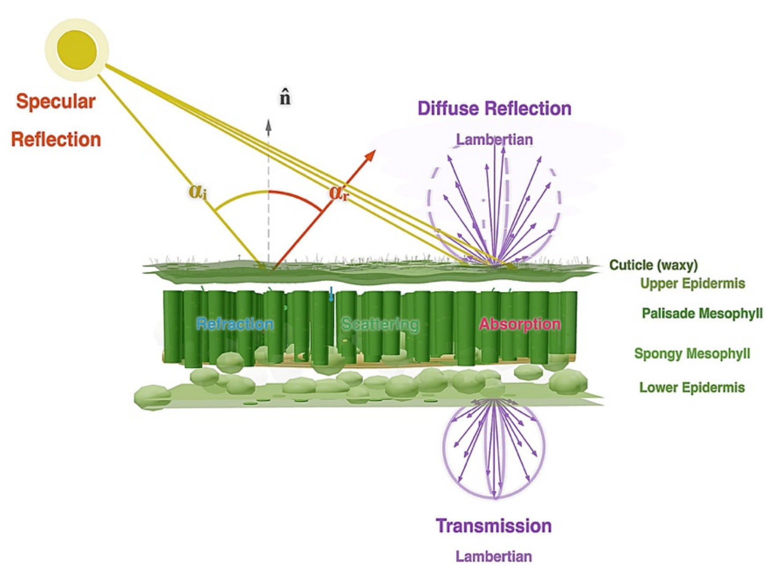
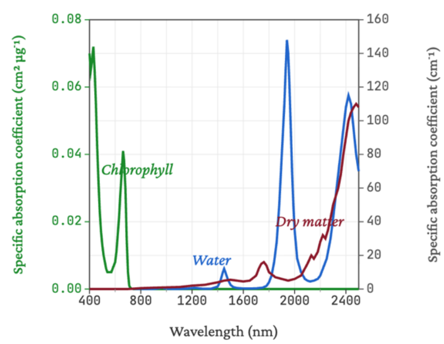


Figure 1. Cross-section of a broadleaf showing the interaction of solar radiation with leaf anatomy.

The cross-section of a conifer needle exhibits a similar layering of tissues. The outermost cuticle is typically thicker and more heavily waxed than on broadleaves, reflecting the xeromorphic adaptations of conifers to reduce water loss. Beneath the epidermis, one or more layers of thick-walled hypodermal cells provide mechanical support. The mesophyll cells, which contain chloroplasts, are arranged more uniformly than in broadleaves, without a clear distinction between palisade and spongy layers. The central transfusion tissue, comprising tracheids and parenchyma cells surrounding the vascular bundle, occupies the interior of the needle. These structural differences contribute to the different optical properties of needles compared with broadleaves: the more uniform mesophyll tends to produce a more nearly symmetric ratio of reflectance to transmittance.

The biochemical constituents determine a leaf's spectral absorption. A typical cell of a fresh leaf contains water (90–95% by volume, contained in the vacuole) and dry matter (5–10%), the latter comprising cellulose (15–30% of dry mass), hemicellulose (10–30%), proteins (10–20%), lignin (5–15%), and smaller amounts of starch, sugars, and other compounds. The pigments include chlorophyll *a* and *b* (contained in the chloroplasts, with area-based concentrations typically 10–100 $\mu g\ cm^{-2}$), the accessory carotenoids (carotenes and xanthophylls), anthocyanins (responsible for red and purple colouration), and brown pigments associated with senescence.

Figure 2. Specific absorption coefficient of chlorophyll a+b ($cm^2\ \mu g^{-1}$), water (cm^{-1}) and dry matter ($cm^2\ g^{-1}$) on the right scale (Jacquemoud, Stéphane & Ustin, Susan. (2003). Application of radiative transfer models to moisture content estimation and burned land mapping. In Proc. 4th International Workshop on Remote Sensing and GIS Applications to Forest Fire Management (E. Chuvieco, P. Martin and C. Justice, eds), Ghent (Belgium), 5-7 June 2003, pp. 3-12).



The absorption spectra of the leaf pigments impose the characteristic spectral signature that makes vegetation distinguishable from all other terrestrial surface types (Fig. 2). Chlorophyll *a* exhibits absorption maxima at 430 nm and 662 nm; chlorophyll *b* at 453 nm and 642 nm. Together, the chlorophylls produce strong absorption throughout the blue (400–500 nm) and red (620–690 nm) regions of the visible spectrum, with a relative minimum near 550 nm that gives healthy foliage its green colour. The carotenoids absorb strongly in the blue region (400–500 nm), overlapping with and supplementing chlorophyll absorption. Anthocyanins, when present, absorb in the green region (500–600 nm), producing the red and purple colours of young expanding leaves, senescing foliage, and environmentally stressed vegetation.

In the near-infrared region (700–1100 nm), pigment absorption is negligible and the leaf's optical properties are governed by internal scattering at the air–cell wall interfaces. The abrupt transition from strong absorption in the red to minimal absorption in the near-infrared—the "red edge" near 700–740 nm—is one of the most distinctive spectral features of vegetation and is widely used in remote sensing for vegetation detection and chlorophyll estimation.

Water absorption features in leaves are found principally around 980 nm and 1240 nm, with very strong absorption bands at 1450 nm and 1950 nm. The shortwave infrared region beyond 1000 nm is dominated by water absorption when leaves are living, and by absorption from the carbon-based cell-wall compounds (cellulose, lignin, hemicellulose) when leaves are dry. Many plant compounds exhibit absorption features in the shortwave infrared, and the cell-wall carbon compounds comprise the largest fraction of leaf dry biomass.

§4. Interactions at the leaf surface and interior

When a collimated beam of radiation is incident upon a leaf, two distinct classes of interaction occur (Fig. 1). At the outermost cuticle surface, a fraction of the radiation is specularly reflected. This surface-reflected component carries no information about the leaf's internal biochemistry; its magnitude depends only on the refractive index of the cuticle and the geometry of illumination and observation. The specular reflection obeys the Fresnel equations and is concentrated about the mirror direction with an angular spread governed by the microscale roughness of the cuticle surface.

The remaining radiation penetrates through the cuticle into the leaf interior, where it is subject to absorption by the pigments and other biochemical constituents, and to scattering at the numerous air–cell wall interfaces within the mesophyll. Photons that survive absorption and emerge from the leaf through the epidermal cells constitute the diffuse component of the leaf's scattered radiation. Those that exit through the illuminated (adaxial) surface contribute to the diffuse reflectance; those that traverse the entire leaf thickness and exit through the opposite (abaxial) surface contribute to the diffuse transmittance. The angular distribution of this diffuse radiation is approximately Lambertian—that is, roughly isotropic in the cosine-weighted sense—because the multiple scattering events within the mesophyll effectively randomise the photon directions.

The total reflectance of a leaf is thus the sum of two components: the specular reflection from the cuticle surface, which is wavelength-independent to first approximation (since the refractive index of cuticular wax varies negligibly across the solar spectrum), and the diffuse reflection from the leaf interior, which carries the full spectral signature of the leaf's biochemistry. Similarly, the transmittance is entirely diffuse, arising from photons that have traversed the leaf interior.

§5. The leaf scattering phase function, albedo, and hemispherical reflectance

Consider a leaf element of area $d\sigma_L$ with outward normal $\mathbf{\Omega}_\mathrm{L}$. The energy dE' (J) in the wavelength interval $d\lambda$ intercepted by this leaf element from intensity I along the direction $\mathbf{\Omega'}$ during the time interval dt is

$$\mathrm{d}E' = I(\mathbf{\Omega'})\,|\mathbf{\Omega'} \cdot \mathbf{\Omega}_\mathrm{L}|\, d\sigma_L\, d\mathbf{\Omega}_\mathrm{L}\, d\lambda\, dt\ .$$

The *leaf scattering phase function* determines dE (J) which is the fraction of intercepted energy that is scattered by the leaf element into a unit solid angle about direction $\mathbf{\Omega}$,

$$\gamma_\mathrm{L}(\mathbf{\Omega'} \rightarrow \mathbf{\Omega};\, \mathbf{\Omega}_\mathrm{L})\, \mathrm{d}\mathbf{\Omega} = dE \,/\, dE'\ . \tag{1}$$

The units of γ_L are sr^{-1}. The scattering phase function is defined over the sphere of outgoing directions, encompassing both reflected radiation (into the hemisphere where $\mathbf{\Omega} \cdot \mathbf{\Omega}_\mathrm{L} > 0$) and transmitted radiation (into the hemisphere where $\mathbf{\Omega} \cdot \mathbf{\Omega}_L < 0$). The dependence on location $\vec{r}$ is suppressed throughout for brevity.

The *leaf albedo* ω is defined as the ratio of total scattered energy to the intercepted energy by the leaf element. It is a dimensionless quantity and includes diffuse reflected and transmitted energy from the leaf interior and specularly reflected energy from the incident surface:

$$\omega(\mathbf{\Omega'}, \mathbf{\Omega}_L) = \int_{4\pi} d\mathbf{\Omega}\, \gamma_L(\mathbf{\Omega'} \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) \,. \tag{2}$$

The *absorptance* is $a_L = 1 - \omega$. The spectral variation of the leaf albedo encodes the leaf's biochemical composition: at 680 nm (~ the chlorophyll absorption maximum), a typical green broadleaf has ~ $\omega \approx 0.1$, reflecting strong pigment absorption; at 740 nm (the near-infrared plateau), ~ $\omega \approx 0.9$, reflecting minimal absorption and copious internal scattering.

The incidence can be on the upper face (adaxial) or the lower face (abaxial) of the leaf. Reflection is defined for cases when $[(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) \times (\mathbf{\Omega} \cdot \mathbf{\Omega}_L)]$ is negative. Transmission is defined when the product is positive. Thus:

$$\rho_L^{-}(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) = \int_{(\mathbf{\Omega}\cdot\mathbf{\Omega}_L)>0} d\mathbf{\Omega}\, \gamma_L(\mathbf{\Omega'} \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) \,, \quad (\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) < 0 \,, \tag{3}$$

$$\tau_L^{-}(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) = \int_{(\mathbf{\Omega}\cdot\mathbf{\Omega}_L)<0} d\mathbf{\Omega}\, \gamma_L(\mathbf{\Omega'} \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) \,, \quad (\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) < 0 \,, \tag{4}$$

$$\rho_L^{+}(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) = \int_{(\mathbf{\Omega}\cdot\mathbf{\Omega}_L)<0} d\mathbf{\Omega}\, \gamma_L(\mathbf{\Omega'} \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) \,, \quad (\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) > 0 \,, \tag{5}$$

$$\tau_L^{+}(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) = \int_{(\mathbf{\Omega}\cdot\mathbf{\Omega}_L)>0} d\mathbf{\Omega}\, \gamma_L(\mathbf{\Omega'} \rightarrow \mathbf{\Omega}; \mathbf{\Omega}_L) \,, \quad (\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) > 0 \,, \tag{6}$$

where ρ_L and τ_L are the *directional-hemispherical reflectance* and *transmittance*. Adaxial and abaxial leaf reflectances and transmittances can be different. However, it is difficult to distinguish adaxial and abaxial incidences in an agglomeration of leaf elements. The scattering coefficient which is defined for an elementary volume containing many leaf elements can be therefore evaluated without much loss of generality by ignoring this variation at individual leaf element scale. Thus,

$$\begin{aligned}
\rho_L^{-}(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) &= \rho_L^{+}(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) = \rho_L(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) \,, \\
\tau_L^{-}(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) &= \tau_L^{+}(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) = \tau_L(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) \,, \\
\omega_L(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) &= \rho_L(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) + \tau_L(\mathbf{\Omega'} \cdot \mathbf{\Omega}_L) \,.
\end{aligned} \tag{7}$$

The reflectance generally exceeds the transmittance at visible wavelengths, primarily because the specular component contributes only to reflectance. At near-infrared wavelengths, the two quantities are comparable in magnitude. A characteristic feature of broadleaves is that the ratio ρ_L/τ_L is typically 2–3 at visible wavelengths and 1.0–1.2 in the near-infrared. This asymmetry arises from two causes: the specular contribution to reflectance, and the structural asymmetry of the palisade and spongy mesophyll layers, whereby the denser palisade layer on the adaxial side preferentially back-scatters radiation toward the illuminated hemisphere. Analogous quantities for a conifer shoot are denoted as ρ_S, τ_S and ω_S.

§6. Diffuse and specular leaf scattering phase functions

Absorption and scattering of radiation by a leaf depend strongly on wavelength. Leaves have a unique spectral signature comprising strong absorption at the red waveband (~680 nm) and strong scattering at the near-infrared waveband (700–1100 nm). This signature is used extensively in remote sensing for vegetation detection and for estimation of properties such as leaf area index.

The angular distribution of radiation reflected by a green leaf shows strong anisotropy, with a characteristic peak about the specular direction. The transmitted radiation, by contrast, is more nearly Lambertian. These angular distributions vary with wavelength, particularly in their magnitude: at 680 nm the reflected hemisphere is dominated by the narrow specular lobe against an otherwise dark background (low diffuse scattering), whereas at 740 nm the diffuse component fills the hemisphere and the specular lobe appears as a modest directional enhancement.

It is convenient to express the leaf scattering phase function as the sum of its specular and diffuse components:

$$\gamma_{L}(\mathbf{\Omega}' \to \mathbf{\Omega}; \mathbf{\Omega}_{L}) = \gamma_{LS}(\mathbf{\Omega}' \to \mathbf{\Omega}; \mathbf{\Omega}_{L}) + \gamma_{LD}(\mathbf{\Omega}' \to \mathbf{\Omega}; \mathbf{\Omega}_{L}) \,, \tag{8}$$

where γ_{LS} describes specular reflection from the cuticle surface and γ_{LD} describes diffuse scattering from the leaf interior. This decomposition is physically motivated by the distinct origins of the two components: the specular component arises at the outermost cuticle surface and is to first approximation wavelength-independent, depending only on the refractive index and surface roughness; the diffuse component arises from multiple scattering within the mesophyll and carries the full spectral signature of the leaf's biochemistry. Simple models can then be formulated independently for each component.

§7. Diffuse interactions: the PROSPECT leaf optical properties model.

The PROSPECT model (Jacquemoud and Baret, 1990) simulates the directional–hemispherical reflectance and transmittance of plant leaves—that is, the magnitudes of the diffuse component—as a function of leaf biochemical content and anatomical structure. PROSPECT (*Pr*opriétés *Opt*iques *Spect*rales des feuilles) is built upon a generalised plate model which treats the leaf as a stack of N homogeneous absorbing plates separated by $N - 1$ air spaces that act as Lambertian scattering interfaces. The model combines transmission through an elementary layer using the Beer–Lambert law with multiple reflection between N layers via the Stokes adding–doubling formalism.

Inputs to the PROSPECT leaf model include the structure parameter N (the number of elementary layers; typically $N \approx 1.0$–2.5 for broadleaves) and the area-based concentrations of the spectrally active biochemical constituents. In more recent versions, these include chlorophyll a+b concentration (typically 30 μg cm^{-2}), carotenoids (10 μg cm^{-2}), anthocyanins (1 μg cm^{-2}), equivalent water thickness (0.09 g cm^{-2}), and dry matter content (0.0015 g cm^{-2}). The absorption coefficient at each wavelength is the sum of contributions from these constituents:

$$k(\lambda) = \Sigma\, C_i K_i(\lambda) \,, \tag{9}$$

where C_i is the content (mass per unit leaf area) of the i-th constituent and $K_i(\lambda)$ its specific absorption coefficient. The model outputs leaf reflectance and transmittance spectra in the wavelength range 400–2500 nm. These outputs are directional–hemispherical quantities representing the result of photon transport and interactions in the interior of the leaf; they do not include the specular component reflected from the cuticle surface, which must be modelled separately.

A sensitivity analysis of PROSPECT reveals the relative contributions of each constituent to the leaf reflectance spectrum. Chlorophyll dominates the signature in the visible region (400–700 nm), the structure parameter N contributes most in the near-infrared (700–1100 nm) through its control over the number of scattering interfaces, and water absorption features in the shortwave infrared (beyond 1400 nm) impart a characteristic signature. The contribution of dry matter is generally less than 15% and becomes important only at wavelengths exceeding 1500 nm. PROSPECT has undergone continuous refinement over three decades. Despite this evolution, the physical framework—plate model, Beer–Lambert absorption, Stokes adding–doubling—has remained essentially unchanged.

§8. Diffuse interactions: the bi-Lambertian model

A fraction ρ_{LD} of incident energy is assumed reflected in a cosine distribution (Lambertian) about the leaf normal. Similarly, another fraction τ_{LD} of incident energy is assumed transmitted in a cosine distribution on the opposite side of the leaf (Fig. 3).

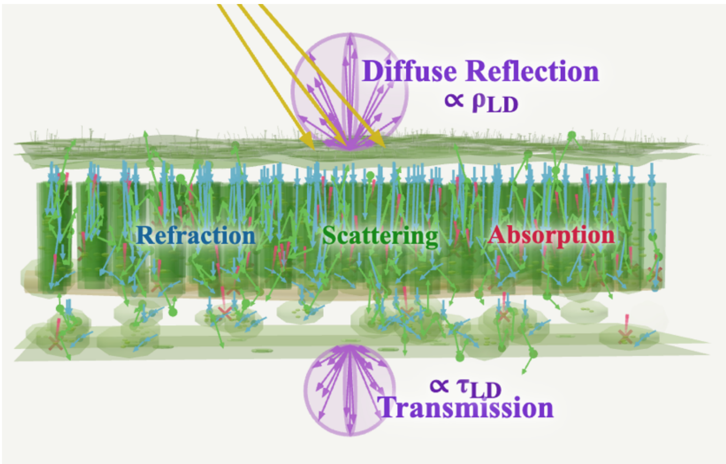


Figure 3. The bi-Lambertian model of diffuse leaf scattering. Incoming sunlight undergoes multiple scattering at air–cell wall interfaces, refraction at cell boundaries, and absorption by chloroplast pigments. Photons that survive absorption and exit produce diffuse reflectance and transmittance.

This *bi-Lambertian* model, originally proposed by Ross (1981), can be expressed as,

$$\gamma_{\mathrm{LD}}(\mathbf{\Omega}' \to \mathbf{\Omega};\ \mathbf{\Omega}_{\mathrm{L}}) = \begin{cases} \frac{1}{\pi}\, \rho_{\mathrm{LD}}\, |\mathbf{\Omega} \cdot \mathbf{\Omega}_{\mathrm{L}}|, & (\mathbf{\Omega} \cdot \mathbf{\Omega}_{\mathrm{L}})(\mathbf{\Omega}' \cdot \mathbf{\Omega}_{\mathrm{L}}) < 0\,, \\ \frac{1}{\pi}\, \tau_{\mathrm{LD}}\, |\mathbf{\Omega} \cdot \mathbf{\Omega}_{\mathrm{L}}|, & (\mathbf{\Omega} \cdot \mathbf{\Omega}_{\mathrm{L}})(\mathbf{\Omega}' \cdot \mathbf{\Omega}_{\mathrm{L}}) > 0\,. \end{cases} \tag{10}$$

It is generally assumed without much loss of generality, and due to lack of data, that ρ_{LD} and τ_{LD} are independent of leaf normal orientation $\mathbf{\Omega}_{\mathrm{L}}$, incident photon direction $\mathbf{\Omega}'$ and location. However, ρ_{LD} and τ_{LD} depend on wavelength of the incident photon and are typically obtained from either measurements or models such as PROSPECT. It follows that $\omega_{\mathrm{LD}} = \rho_{\mathrm{LD}} + \tau_{\mathrm{LD}}$ is the leaf albedo of diffuse interactions inside the leaf. The combination of PROSPECT (for the spectral magnitudes) and the bi-Lambertian model (for the angular distribution) provides a complete description of the diffuse component γ_{LD} of the leaf scattering phase function.

§9. Surface interactions: the Cook–Torrance specular reflection model

The specular component of the leaf scattering phase function was originally formulated using a two-dimensional Dirac delta function:

$$\gamma_{\mathrm{LS}}(\mathbf{\Omega'} \rightarrow \mathbf{\Omega};\, \mathbf{\Omega}_{\mathrm{L}}) = \delta_2(\mathbf{\Omega} \cdot \mathbf{\Omega}^*)\, F_{\mathrm{r}}(n, \alpha')\, K(\kappa, \alpha')\,, \tag{11}$$

where δ_2 constrains scattering to the specular direction $\mathbf{\Omega}^*$ (the mirror reflection of $\mathbf{\Omega'}$ about the leaf normal), F_{r} is the Fresnel reflectance, and K is a roughness attenuation factor. The Dirac delta idealises the leaf surface as perfectly smooth, concentrating all reflected energy into an infinitesimally narrow specular spike.

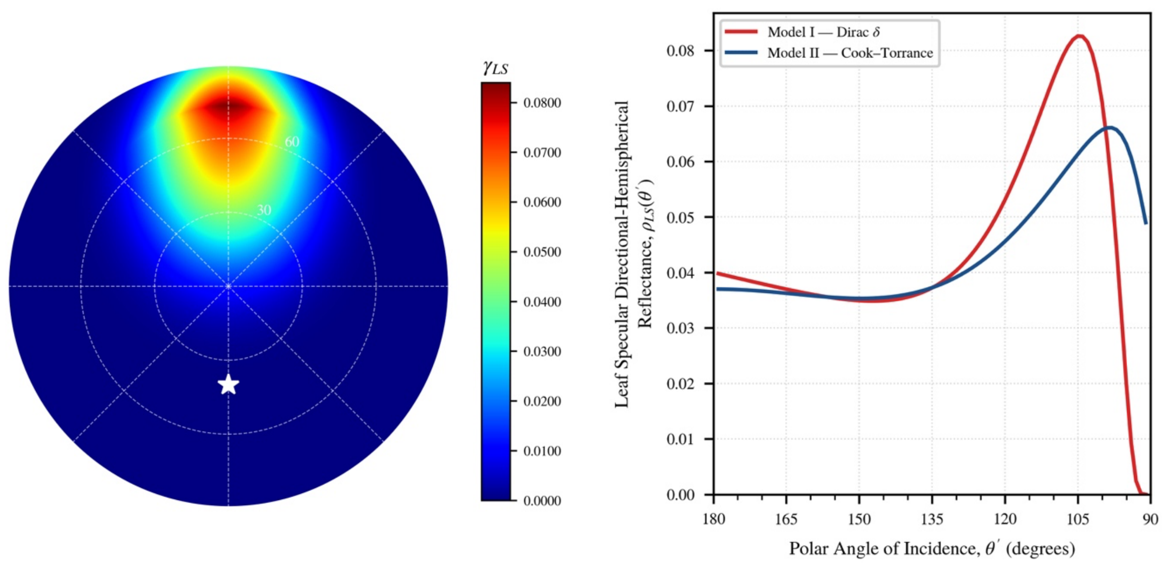


Figure 5. Left panel: polar plot of the broadleaf specular scattering phase function $\gamma_{\mathrm{LS}}(\mathbf{\Omega'} \rightarrow \mathbf{\Omega};\, \mathbf{\Omega}_{\mathrm{L}})$ for a horizontal leaf computed from the Cook–Torrance microfacet model with Beckmann roughness parameter $m = 0.3$, refractive index $n = 1.5$ and extinction coefficient $\kappa = 0.3$. The incident direction is $\mathbf{\Omega'} = (\theta', \varphi') = (140°, 0°)$; the specular lobe peaks slightly beyond the geometric mirror direction owing to the masking–shadowing function. Right panel Directional–hemispherical specular reflectance ρ_{LS} as a function of the polar angle of incidence $\theta' \in (90°, 180°]$ for a horizontal leaf computed with refractive index $n = 1.5$. Left axis (blue): Cook–Torrance microfacet model with Beckmann roughness $m = 0.30$. At normal incidence ($\theta' = 180°$) both converge to the Fresnel reflectance at normal incidence, $F_{\mathrm{r}}(1, n) = 0.040$. Right axis (red): Dirac delta model with roughness extinction factor $\kappa = 0.3$.